Questions

Q1.

Radioactivity is used in PET scanners in hospitals.

(i) Describe **one** use of PET scanners in hospitals.

(2)

.....

.....

.....

.....

(ii) State **two** precautions that hospital staff should take when working with radioactivity.

(2)

1 .....

.....

2 .....

.....

(Total for question = 4 marks)

Q2.

Figure 7 is a diagram of three atoms.

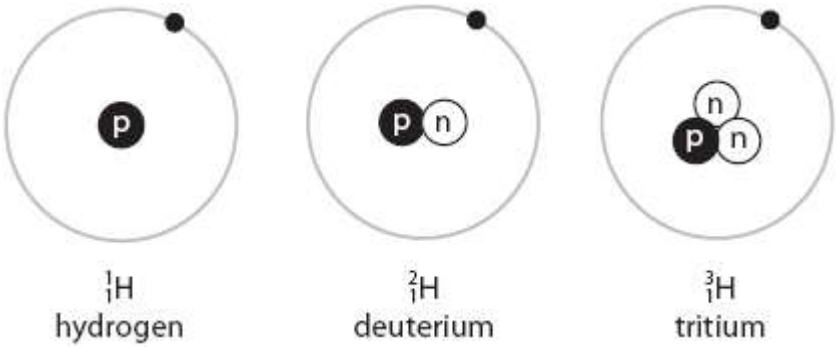

Figure 7

Give reasons why these atoms are isotopes.

(2)

.....

.....

.....

.....

(Total for question = 2 marks)

Q3.

Medical staff who use radioactive materials need more protection than their patients.

Describe some precautions that medical staff can take to ensure their safety from radioactive materials.

(3)

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Q4.

Explain how radiation from radioactive sources can be dangerous to people.

(2)

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.....

.....

Q5.

Many different types of radiation are used by doctors.

Which type of radiation comes from radioactive sources?

Put a cross (☒) in the box next to your answer.

(1)

☐ **A** gamma rays

☐ **B** ultrasound

☐ **C** ultraviolet

☐ **D** X-rays

Q6.

Choose words from the box to complete the following sentences.

Words may be used once, more than once or not at all.

alpha	beta	gamma	positron

The radiation that is a wave is .....

(1)

The particle that is negatively charged is .....

(1)

Q7.

A student uses 59 dice to model radioactive decay.

He starts by rolling all the dice at the same time.

He removes all the dice that show a six.

He then rolls the remaining dice.

The student repeats this process five more times.

State **two** improvements the student could make to his model of radioactive decay.

(2)

1 .....

.....

2 .....

.....

(Total for question = 2 marks)

Q8.

Which of the following particles is absorbed and emitted during a nuclear fission reaction?

(1)

- ☐ **A** electron
- ☐ **B** neutron
- ☐ **C** positron
- ☐ **D** proton

(Total for question = 1 mark)

Q9.

Other unstable isotopes emit alpha particles.

Which of these describes an alpha particle?

(1)

- ☐ **A** a hydrogen nucleus
- ☐ **B** a hydrogen atom
- ☐ **C** a helium nucleus
- ☐ **D** a helium atom

(Total for question = 1 mark)

Q10.

Radiation is used to treat tumours (cancer).

The source of the radiation can be inside or outside the human body.

Which of these has a radiation source that can be positioned inside the body to treat tumours?

(1)

- ☐ **A** gamma rays
- ☐ **B** x-rays
- ☐ **C** radio waves
- ☐ **D** microwaves

(Total for question = 1 mark)

Mark Scheme

Q1.

Question Number	Answer	Additional guidance	Mark
(i)	A description to include: diagnosing / locating / identify/look for (1) tumour (1)	brain scan cancer award 2 marks for 'seeing where the cancer is'	(2)

Question Number	Answer	Additional guidance	Mark
(ii)	Any two precautions to do with:		(2)
	mp1 short half-life isotopes (1)		
	mp2 shielding (1)	protective clothing lead shielding	
	mp3 distance (1)	working in another room	
	mp4 isolation (1)	working	
	mp5 exposure time (1)	radiation badges	
	mp6 transportation (1)	to and around the hospital	
	mp7 storage (1)	appropriate containers	

Q2.

Question number	Answer	Mark
	one from • same atomic number (1) • same number of protons (1) • same element (1) and one from • different numbers of neutrons (1) • different mass numbers (1)	(2)

Q3.

Question Number	Answer	Acceptable answers	Mark
	Any three from:		**(3)**
	• keep distant from sources / (stand) in a separate room (behind leaded window etc.)	(distance also involved if you) use computer controlled equipment	
	• limit **time** exposed to the radioactivity	the time aspect must be clear here.	
	• use lead shielding for the sources / handle sources with tongs etc. / dispose radioactive material(s) safely		
	• wear lead aprons / used lead-lined clothing / lead-lined gloves	ignore goggles / (special) gloves without detail. Similarly ignore 'radiation resistant' (clothes)	
	• monitor exposure with some detector / badge / use of (radiation) meters		

Q4.

Question Number	Answer	Acceptable answers	Mark
	An explanation linking the following:-		**(2)**
	(it is) ionising (can cause) (1)	has high frequency/energy	
	damage to tissue/ mutation/cancer/tumours (1)	kill cells / (causes) burns	

Q5.

Question Number	Answer	Acceptable answers	Mark
	A gamma		**(1)**

Q6.

Question Number	Answer	Acceptable answers	Mark
	gamma (1) beta (1)		**(2)**

Q7.

Question number	**Answer**	**Mark**
	• increase number of starting dice (1) • do more rolls (1)	**(2)**

Q8.

Question number	**Answer**	**Mark**
	B neutron	**(1)**

Q9.

Question number	**Answer**	**Mark**
	C a helium nucleus	**(1)**

Q10.

Question	Answer	Mark
	A gamma rays **B** X-rays, **C** ultraviolet, **D** microwaves, all applied externally	**1** **AO1.1**